Unit-6: Logistic Regression

1. Why Logistic Regression is Needed

Linear regression fails when output is **categorical**.

Example Problem

Input	Output
Study hours	Pass / Fail
Email text	Spam / Not Spam
Patient data	Disease / No Disease

☞ Output is **not numeric continuous** → it is **classification**

2. Type of Problems

Problem Type	Output
Regression	continuous (price, marks)
Classification	categorical (yes/no, 0/1)

☞ Logistic Regression is used for **classification problems**

3. Key Idea of Logistic Regression

Instead of predicting value directly:

Predict Probability of an event

Example:

P(Pass) = 0.8 → student likely to pass

P(Pass) = 0.2 → student likely to fail

4. Important Terms You Must Know

1. Binary Classification

Definition: When output has **only two classes**

Example
Yes / No
0 / 1
True / False

☞ Logistic Regression is mainly used for **binary classification**

2. Class Labels

Label	Meaning
0	Negative class
1	Positive class

Example:

Problem	0	1
Email	Not Spam	Spam
Exam	Fail	Pass

3. Probability

Definition: A value between: $0 \le P \le 1$

Interpretation

Probability	Meaning

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0	impossible
0.5	uncertain
1	certain

4. Odds
Definition: Odds represent **ratio of success to failure**
Odds = P / (1 − P)

Example
If:
P = 0.8
Then:
Odds = 0.8 / 0.2 = 4
☞ Success is 4 times more likely

5. Log-Odds (Logit Function)
Definition
Log of odds:
log(p / (1 − p))
☞ This is called **logit**

Why Important
- Converts probability → real number
- Makes model **linear**

6. Sigmoid Function (MOST IMPORTANT)
Definition: A function that converts any value into range (0,1)
σ(z) = 1 / (1 + e^(−z))

Output Shape

Input (z)	Output
very large +ve	≈ 1
0	0.5
very large −ve	≈ 0

☞ This function is the **core of Logistic Regression**

7. Decision Boundary
Definition: A threshold used to classify output

Example
If P ≥ 0.5 → Class 1
If P < 0.5 → Class 0

8. Independent and Dependent Variable
Same as regression:

Type	Meaning
X	input
Y	output (0 or 1)

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9. Linear Combination
Before applying sigmoid:
$z = b_0 + b_1X$
Then:
P = sigmoid(z)

10. Classification Output
Final output:

Probability	Class
0.7	1
0.3	0

5. Difference: Linear vs Logistic Regression

Feature	Linear Regression	Logistic Regression
Output	continuous	categorical
Range	−∞ to +∞	0 to 1
Function	straight line	sigmoid curve
Use	prediction	classification

6. Intuition (Very Important)
Input (X)
↓
Linear equation ($z = b_0 + b_1X$)
↓
Apply sigmoid
↓
Get probability (0 to 1)
↓
Apply threshold
↓
Final class (0 or 1)

7. Real-Life Examples

Domain	Example
Email filtering	spam detection
Healthcare	disease prediction
Finance	loan approval
E-commerce	customer churn

8. Why Logistic Regression is Important for You
For a **Data Scientist role (Oracle, etc.)**, this is:

- basic classification algorithm
- used in real production systems
- foundation for advanced ML

Introduction – Classification Problems

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Before learning Logistic Regression, you must clearly understand **classification problems**, because Logistic Regression is designed specifically to solve them.

1. What is a Classification Problem

Definition: A **classification problem** is a type of problem where the output variable is **categorical (discrete)** rather than continuous.

☞ Instead of predicting a number, we predict a **category or class label**

Example

Input (X)	Output (Y)
Email content	Spam / Not Spam
Study hours	Pass / Fail
Medical data	Disease / No Disease

2. Key Difference: Regression vs Classification

Feature	Regression	Classification
Output type	continuous	categorical
Example	price prediction	spam detection
Algorithm	linear regression	logistic regression

3. Types of Classification Problems

1. Binary Classification (Most Important)

Definition: Only **two possible classes**

Examples
Yes / No
0 / 1
True / False

Example Table

Feature	Class
Study hours	Pass (1) / Fail (0)

☞ Logistic Regression mainly focuses on this

2. Multi-Class Classification

Definition: More than two classes

Example
Grade: A, B, C
Traffic signal: Red, Yellow, Green

3. Multi-Label Classification

Definition: One input can belong to **multiple classes**

Example:

Input	Output
Movie	Action + Comedy

4. Understanding Classes and Labels

Term	Meaning
Class	category of output

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Label	numeric representation

Example

Class	Label
Fail	0
Pass	1

5. Decision Boundary (Core Concept)

Definition

A **decision boundary** is a rule used to separate classes.

Example

If $P \geq 0.5 \rightarrow$ Class 1

If $P < 0.5 \rightarrow$ Class 0

👉 This threshold divides the data into categories

6. Why Linear Regression Fails for Classification

Problem 1: Output Range

Linear regression output:

$-\infty$ to $+\infty$

But classification requires:

0 to 1

Problem 2: No Probability Meaning

Linear regression does not provide:

- probability
- class confidence

Problem 3: Poor Decision Boundary

Linear regression may produce:

- incorrect classification
- unstable results

7. Solution: Logistic Regression

To solve classification problems, we:

1. Use linear equation
2. Apply sigmoid function
3. Convert output into probability

Flow

Input (X)

↓

Linear model ($z = b_0 + b_1X$)

↓

Sigmoid function

↓

Probability (0 to 1)

↓

Apply threshold

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↓
Final class (0 or 1)

8. Example (Clear Understanding)
Problem: Predict if student will pass

Study Hours	Pass/Fail
2	Fail (0)
4	Fail (0)
6	Pass (1)
8	Pass (1)

👉 Here:
- Output is categorical
- Cannot use linear regression
- Logistic regression is suitable

9. Applications of Classification Problems

Domain	Example
Email systems	spam detection
Healthcare	disease prediction
Banking	loan approval
E-commerce	customer churn

10. Important Concepts in Classification

Concept	Meaning
Class	category
Label	numeric form
Probability	chance of class
Threshold	decision rule
Prediction	final class

11. Real-World Thinking
Classification answers questions like:
- Will customer buy or not?
- Will patient survive or not?
- Is transaction fraud or not?

👉 These are **decision-making problems**

12. Concept Flow (Very Important)
Real-world problem
↓
Categorical outcome
↓
Define classes (0,1)
↓
Use model (Logistic Regression)
↓
Predict probability

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↓
Convert to class

Introduction to Binary Logistic Regression

This is the **core concept of Unit-6**. It explains how we move from a simple linear model to a **probability-based classification model**.

1. What is Binary Logistic Regression

Definition: Binary Logistic Regression is a statistical method used to model the relationship between input variables (X) and a binary outcome ($Y \in \{0,1\}$) using probability.

Key Idea

Instead of predicting a number, we predict:
Probability of belonging to class 1

Example

Input (X)	Output (Y)
Study hours	Pass (1) / Fail (0)

☞ Output is **binary (0 or 1)**

2. Why Not Use Linear Regression

If we apply linear regression:
$Y = b_0 + b_1X$
Problems:

- Output can be < 0 or > 1 ✕
- No probability meaning ✕
- Poor classification ✕

☞ Solution: Use **Logistic Function (Sigmoid)**

3. Core Idea of Logistic Regression

Step 1: Linear Model

$z = b_0 + b_1X$

Step 2: Apply Sigmoid Function

$P(Y=1) = 1 / (1 + e^{(-z)})$

Step 3: Classification

If $P \geq 0.5 \rightarrow$ Class 1
If $P < 0.5 \rightarrow$ Class 0

4. Sigmoid Function (Heart of Logistic Regression)

Formula

$\sigma(z) = 1 / (1 + e^{(-z)})$

Properties

Input z	Output σ(z)
Large positive	≈ 1

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0	0.5
Large negative	≈ 0

Meaning

- Converts any number → range (0,1)
- Makes output a **probability**

5. Graph Intuition

Input (z) → Sigmoid Curve → Probability (0 to 1)

- S-shaped curve
- Smooth transition between classes

6. Logistic Regression Equation

Probability Form

$P(Y=1) = 1 / (1 + e^{(-(b_0 + b_1X))})$

Logit Form (Linear Form)

$\log(p / (1 - p)) = b_0 + b_1X$

Meaning

Form	Purpose
Probability	prediction
Logit	linear modeling

7. Understanding Log-Odds (Logit)

Step 1: Odds

Odds = $p / (1 - p)$

Step 2: Log-Odds

$\log(p / (1 - p))$

👉 This transforms probability → **linear scale**

8. Decision Boundary

Definition: A threshold used to classify output

Standard Rule

$P \geq 0.5$ → Class 1

$P < 0.5$ → Class 0

Example

Probability	Class
0.8	1
0.3	0

9. Example (Full Understanding)

Suppose:

$z = -4 + 1.5X$

Step 1: Compute z

If X = 4:

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$z = -4 + 1.5(4) = 2$

Step 2: Apply Sigmoid
$P = 1 / (1 + e^{(-2)}) \approx 0.88$

Step 3: Classification
$P = 0.88 \geq 0.5 \rightarrow$ Class 1
👉 Prediction: **Positive class**

10. Interpretation of Coefficients

Term	Meaning
b_0	baseline log-odds
b_1	change in log-odds per unit X

Important Insight
b_1 affects probability indirectly via log-odds

11. Key Differences: Linear vs Logistic

Feature	Linear Regression	Logistic Regression
Output	continuous	probability
Range	$-\infty$ to $+\infty$	0 to 1
Function	straight line	sigmoid curve
Use	prediction	classification

12. Real-World Applications

Domain	Example
Email	spam detection
Healthcare	disease prediction
Banking	loan approval
E-commerce	churn prediction

13. Why Logistic Regression is Powerful

Feature	Benefit
probabilistic output	interpretable
simple model	fast computation
strong baseline	widely used
extendable	multi-class models

14. Concept Flow (Very Important)
Input Data (X)
↓
Linear Combination ($z = b_0 + b_1X$)
↓
Apply Sigmoid
↓
Get Probability (0–1)
↓
Apply Threshold

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↓
Final Class (0 or 1)

Estimation of Parameters in Logistic Regression

In linear regression, parameters were estimated using **Least Squares**.
In logistic regression, that approach does **not work** because the output is **probability (non-linear)**.
☞ Therefore, we use a different method:

Maximum Likelihood Estimation (MLE)

1. What Are We Estimating
In binary logistic regression model:

$$P(Y=1) = 1 - \left(1 + e^{-\left(b^0 + b^1 X\right)}\right)$$

We need to estimate:
b_0 → intercept
b_1 → coefficient (slope)

2. Why Least Squares Does NOT Work

Reason	Explanation
non-linear model	sigmoid function is non-linear
output is probability	not continuous
error is not normally distributed	violates assumptions

☞ So we switch to **Likelihood-based estimation**

3. Concept of Maximum Likelihood Estimation (MLE)
Idea: Choose parameters that maximize the probability of observing the given data

Meaning Find b_0, b_1 such that observed data is most likely

4. Likelihood Function
For binary classification:

Y	Probability
Y = 1	P
Y = 0	(1 − P)

Likelihood for Single Observation

$$L = P^Y \times (1 - P)^{1-Y}$$

For All Observations

$$L = \Pi \left[P_{i_i^Y} \times (1 - P_i)^{1-Y_i}\right]$$

5. Log-Likelihood Function
To simplify computation, we take log:
log L = Σ [Y log(P) + (1 − Y) log(1 − P)]

☞ This is the function we maximize

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6. Objective of Logistic Regression
Maximize log-likelihood
☞ Find values of **b_0 and b_1** that maximize this

7. How Optimization is Done
Unlike linear regression:

- No direct formula ✕
- Uses **iterative methods** ✔

Common Methods

Method	Description
Gradient Descent	step-by-step optimization
Newton-Raphson	faster convergence
Iterative algorithms	repeated updates

8. Gradient Descent Intuition
Idea : Start with random values → improve step by step

Process
Initialize b_0, b_1
↓
Compute error (log-likelihood)
↓
Update parameters
↓
Repeat until convergence

9. Cost Function (Loss Function)
Instead of SSE, we use:
Log Loss (Cross-Entropy Loss)
Loss = − Σ [Y log(P) + (1 − Y) log(1 − P)]

Interpretation

Prediction	Loss
correct prediction	low loss
wrong prediction	high loss

10. Example (Simple Intuition)
Suppose:

X	Y
2	0
4	1

Model predicts:
P = 0.8 for Y=1
If actual Y = 1 → good prediction ✔
If actual Y = 0 → bad prediction ✕
☞ Loss function penalizes wrong predictions

11. Output After Estimation

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After optimization, we get:
b_0 = value
b_1 = value
Final model:

$$P(Y=1) = 1 - \left(1 + e^{-(b^0 + b^1 X)}\right)$$

12. Interpretation of Parameters

Parameter	Meaning
b_0	baseline log-odds
b_1	effect of X on log-odds

Important Insight
b_1 changes probability through log-odds

13. Why MLE is Important

Feature	Benefit
works for non-linear models	suitable for sigmoid
probabilistic approach	meaningful interpretation
flexible	used in ML models

14. Real-World Example
Predict loan approval:

Input	Output
income, credit score	approved (1) / not (0)

MLE finds parameters that make observed approvals most likely.

15. Concept Flow (Very Important)
Data (X, Y)
↓
Apply Logistic Model
↓
Compute Probability (P)
↓
Construct Likelihood Function
↓
Take Log-Likelihood
↓
Maximize Function
↓
Get b_0, b_1

16. Key Summary

Concept	Meaning
MLE	maximize likelihood
Likelihood	probability of data
Log-likelihood	simplified form
Loss function	cross-entropy
Optimization	iterative methods

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17. Important Exam Points

Focus on:

- why least squares not used
- likelihood function formula
- log-likelihood formula
- concept of gradient descent
- cross-entropy loss

Interpretation of Logistic Regression Parameters

This topic explains how to **understand the meaning of coefficients (b_0, b_1, ...)** in logistic regression and how they affect probability.

1. What Are Parameters in Logistic Regression

In logistic regression, the model is:

$$P(Y = 1) = \frac{1}{1 + e - (b0 \quad + b1 \quad X)}$$

OR in logit form:

Parameters:

Parameter	Meaning
b_0	Intercept (baseline)
b_1	Effect of input variable X

2. Key Idea of Interpretation

☞ Logistic regression coefficients are interpreted in terms of:

- **Log-odds (NOT direct probability)**
- **Change in odds**

3. Interpretation of Intercept (b_0)

Definition: Intercept represents **log-odds when X = 0**

Meaning:

- Baseline probability of outcome when no input is present

Example:

If:

$b_0 = -2$

Then:

Log-odds = -2

Convert to probability:

P ≈ 0.12

$$\log\left(\frac{p}{1-p}\right) = b_0 + b_1 X$$

☞ Without any input, probability of event is low

4. Interpretation of Coefficient (b_1)

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Definition:
b_1 represents **change in log-odds for 1 unit increase in X**

Important Insight:

Increase in X	Effect
+1 unit	log-odds change by b_1

Case 1: $b_1 > 0$
- Increases probability
- Positive relationship

☞ Example:
More study hours → higher chance of passing

Case 2: $b_1 < 0$
- Decreases probability
- Negative relationship

☞ Example:
More debt → lower chance of loan approval

Case 3: $b_1 = 0$
- No effect of X
- Probability remains same

5. Odds Interpretation (Very Important)
Instead of log-odds, we use **odds ratio**
Formula:
Odds Ratio = $e^{(b_1)}$

Meaning:

Odds Ratio	Interpretation
> 1	Increase in odds
< 1	Decrease in odds
= 1	No effect

Example:
If:
$b_1 = 0.7$
Then:
Odds Ratio = $e^{(0.7)} \approx 2$
☞ Odds increase **2 times** for each unit increase in X

6. Example (Clear Understanding)
Model:
$z = -4 + 1.5X$

Step 1: Interpret b_0
$b_0 = -4$
☞ When X = 0 → log-odds = -4 → very low probability

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Step 2: Interpret b_1

$b_1 = 1.5$

☞ Each 1 unit increase in X:

- log-odds increase by 1.5
- odds multiply by e^(1.5) ≈ 4.48

☞ Strong positive effect

7. Multiple Variable Case

Model:

$$\log\left(\frac{p}{1-p}\right) = b_0 + b_1X_1 + b_2X_2$$

Interpretation:

Parameter	Meaning
b_1	effect of X_1 (keeping X_2 constant)
b_2	effect of X_2 (keeping X_1 constant)

☞ Each variable is interpreted **independently**

8. Probability Interpretation (Important)

⚠ Important:

- Coefficients do NOT directly change probability
- They change **log-odds → which indirectly changes probability**

Insight:

Small change in X → nonlinear change in probability

☞ Because of sigmoid curve

9. Summary Table

Parameter	Interpretation
b_0	baseline log-odds
b_1	change in log-odds per unit X
e^(b_1)	change in odds
sign of b_1	direction of relationship

10. Key Points for Exam

- Coefficients are interpreted in **log-odds scale**
- Use **odds ratio** $\left(e^b\right)$ for better understanding
- Positive coefficient → increases probability
- Negative coefficient → decreases probability
- Intercept = baseline when X = 0
- Interpretation is **not linear in probability**

11. Concept Flow (Very Important)

Input variable (X)

↓

Affects coefficient (b_1)

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↓
Changes log-odds
↓
Changes odds
↓
Changes probability
↓
Final classification

Logistic Regression Model Diagnostics

This topic is directly mentioned in your syllabus. It focuses on **checking how well the logistic regression model fits and performs**.

1. What is Model Diagnostics

Definition: Model diagnostics is the process of **evaluating the performance and reliability of a logistic regression model**.

☞ It answers:

- Is the model correct?
- Is it predicting well?
- Can we trust the results?

2. Why Model Diagnostics is Needed

Problem	Explanation
Poor predictions	Model may misclassify data
Overfitting	Model fits training data too much
Underfitting	Model is too simple
Wrong variables	Irrelevant features included

☞ Diagnostics helps improve model quality

3. Types of Diagnostics

Type	Purpose
Goodness of Fit	How well model fits data
Classification Accuracy	How correct predictions are
Residual Analysis	Detect errors/outliers
Statistical Tests	Validate model assumptions

4. Goodness of Fit (Very Important)

Idea: Check how well model explains the data

4.1 Likelihood Concept

- Logistic regression uses **Likelihood**
- Higher likelihood → better model

4.2 Deviance

Formula Idea:

Deviance = measure of error

Value	Meaning
Low deviance	Good model

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High deviance	Poor model

4.3 Null vs Full Model

Model	Meaning
Null Model	No predictors
Full Model	With predictors

☞ Compare both:
If Full Model is better → variables are useful

4.4 Pseudo R^2 (Important)
Similar to R^2 in linear regression
Types:

- Cox & Snell R^2
- Nagelkerke R^2

Value	Meaning
Close to 1	Good model
Close to 0	Poor model

5. Classification Accuracy
Definition: Measures how correctly model classifies data

Formula: Accuracy = (TP + TN) / Total

Example:

Actual	Predicted
1	1 ✔
0	0 ✔
1	0 ✕

☞ More correct predictions → higher accuracy

6. Confusion Matrix (Core Tool)

Actual \ Predicted	1	0
1 (Positive)	TP	FN
0 (Negative)	FP	TN

Terms:

Term	Meaning
TP	Correct positive
TN	Correct negative
FP	Wrong positive
FN	Wrong negative

7. Residual Analysis
Definition:
Residual = difference between actual and predicted

Types of Residuals:

Type	Purpose

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Pearson Residual	detect large errors
Deviance Residual	measure model fit

Use:

- Identify outliers
- Detect model issues

8. Hosmer–Leme show Test (Important)

Purpose:

Check goodness of fit

Idea:

Compare:

- observed values
- predicted values

Result:

p-value	Meaning
> 0.05	Good fit ✔
< 0.05	Poor fit ✕

9. Multicollinearity Check

Problem: When independent variables are highly correlated

Effect:

- Unstable coefficients
- Wrong interpretation

Detection:

Method	Meaning
VIF (Variance Inflation Factor)	High value = problem

10. Overfitting and Underfitting

Case	Meaning
Overfitting	fits training data too well
Underfitting	too simple model

Solution:

Problem	Solution
Overfitting	regularization
Underfitting	add variables

11. Key Summary Table

Concept	Meaning
Goodness of fit	how well model fits
Deviance	model error
Pseudo R^2	model strength
Confusion matrix	prediction evaluation
Residuals	error detection

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Hosmer-Lemeshow	fit test
VIF	multicollinearity

12. Concept Flow (Very Important)
Model Built
↓
Check Likelihood & Deviance
↓
Evaluate Accuracy
↓
Use Confusion Matrix
↓
Analyze Residuals
↓
Apply Statistical Tests
↓
Final Model Validation

Classification Table (Confusion Matrix)

This topic is part of Unit-6 (Logistic Regression) in your syllabus
It is used to **evaluate how well the model classifies outcomes**.

1. What is Classification Table
Definition: A Classification Table (also called **Confusion Matrix**) is a table used to **compare actual values with predicted values**.
☞ It shows:

- Correct predictions
- Incorrect predictions

2. Structure of Classification Table

Actual \ Predicted	1 (Positive)	0 (Negative)
1 (Positive)	TP	FN
0 (Negative)	FP	TN

3. Important Terms

Term	Full Form	Meaning
TP	True Positive	Correctly predicted positive
TN	True Negative	Correctly predicted negative
FP	False Positive	Wrongly predicted positive
FN	False Negative	Wrongly predicted negative

4. Understanding with Example
Problem: Disease Prediction

Actual	Predicted
Disease	Disease ✔
Disease	No Disease ✕
No Disease	Disease ✕
No Disease	No Disease ✔

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Mapping:

Case	Type
Disease → Disease	TP
Disease → No Disease	FN
No Disease → Disease	FP
No Disease → No Disease	TN

5. Why Classification Table is Important

Purpose	Explanation
Evaluate model	Check performance
Identify errors	FP and FN
Compare models	Choose best model

6. Accuracy (Basic Measure)

Formula:

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Meaning:

Value	Interpretation
High accuracy	Good model
Low accuracy	Poor model

7. Error Types (Very Important)

1. False Positive (FP)

- Predicts **Yes**, but actually **No**

👉 Example:

Spam filter marks normal email as spam

2. False Negative (FN)

- Predicts **No**, but actually **Yes**

👉 Example:

Disease not detected (dangerous)

8. Insight from Table

Situation	Meaning
High TP & TN	Good model
High FP	Too many false alarms
High FN	Missing important cases

9. Limitations of Accuracy

⚠ Accuracy alone is not enough

Example:

Total Data	100
Positive cases	5
Negative cases	95

Model predicts all as negative →

Accuracy = 95% ❗ (but useless)

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☞ Need better metrics (next topic)

10. Link to Next Concepts
From classification table, we calculate:

Metric	Formula
Sensitivity	TP / (TP + FN)
Specificity	TN / (TN + FP)

☞ These are more reliable than accuracy

11. Summary Table

Concept	Meaning
Classification Table	compares actual vs predicted
TP	correct positive
TN	correct negative
FP	false alarm
FN	missed case
Accuracy	overall correctness

12. Concept Flow (Very Important)
Model Prediction
↓
Compare with Actual Values
↓
Fill Confusion Matrix
↓
Calculate Metrics
↓
Evaluate Model Performance

Sensitivity and Specificity

This topic is part of Unit-6 (Logistic Regression) in your syllabus
It is derived from the **classification table (confusion matrix)** and is **very important for exams and viva**.

1. What are Sensitivity and Specificity

Term	Meaning
Sensitivity	Ability to correctly identify positive cases
Specificity	Ability to correctly identify negative cases

☞ These metrics evaluate **model performance more accurately than accuracy**

2. Based on Confusion Matrix

Actual \ Predicted	1	0
1 (Positive)	TP	FN
0 (Negative)	FP	TN

3. Sensitivity (Recall / True Positive Rate)
Definition: Sensitivity measures how well the model identifies **actual positive cases**

Formula: Sensitivity = TP / (TP + FN)

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Interpretation:

Value	Meaning
High sensitivity	Few missed cases (FN low)
Low sensitivity	Many missed cases

Example (Healthcare)

- Detect disease patients
- High sensitivity → fewer patients missed

👉 Very important in medical diagnosis

4. Specificity (True Negative Rate)

Definition : Specificity measures how well the model identifies **actual negative cases**

Formula: Specificity = TN / (TN + FP)

Interpretation:

Value	Meaning
High specificity	Few false alarms (FP low)
Low specificity	Many false alarms

Example (Spam Detection)

- Detect non-spam emails
- High specificity → fewer normal emails marked as spam

5. Key Difference

Feature	Sensitivity	Specificity
Focus	Positive class	Negative class
Measures	Correct positives	Correct negatives
Error reduced	FN	FP

6. Example (Clear Understanding)

Suppose:

Value	Count
TP	40
FN	10
FP	20
TN	30

Calculate:

Sensitivity = 40 / (40 + 10) = 0.8

Specificity = 30 / (30 + 20) = 0.6

Interpretation:

- Sensitivity = 80% → good detection of positives
- Specificity = 60% → moderate detection of negatives

7. When to Focus on Which

Situation	Important Metric

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Disease detection	Sensitivity
Fraud detection	Sensitivity
Spam filtering	Specificity
Quality control	Specificity

8. Trade-off Between Them

☞ Increasing sensitivity may decrease specificity

☞ Increasing specificity may decrease sensitivity

Reason:

Changing **cut-off threshold**

Threshold	Effect
Low threshold	High sensitivity
High threshold	High specificity

9. Link with Cut-Off Probability

Default:

$P \geq 0.5 \rightarrow$ Class 1

But we can change threshold to:

- increase sensitivity
- increase specificity

10. Summary Table

Concept	Formula	Focus
Sensitivity	TP / (TP + FN)	detect positives
Specificity	TN / (TN + FP)	detect negatives

11. Key Exam Points

- Sensitivity = True Positive Rate
- Specificity = True Negative Rate
- Derived from confusion matrix
- Important in **imbalanced datasets**
- Used to evaluate model quality

12. Concept Flow (Very Important)

Confusion Matrix

↓

Calculate TP, TN, FP, FN

↓

Compute Sensitivity

↓

Compute Specificity

↓

Evaluate Model Performance

✔ **Final Understanding**

☞ Accuracy gives overall result

☞ Sensitivity and Specificity give **detailed performance**

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☞ Good model should balance both:

- detect positives correctly
- avoid false alarms

Optimal Cut-Off Probability
This topic is part of Unit-6 (Logistic Regression) in your syllabus
It explains **how to choose the best threshold (cut-off) to convert probability into class labels**.

1. What is Cut-Off Probability
Definition: Cut-off probability is a **threshold value used to classify predicted probability into class labels (0 or 1)**.

Standard Rule:

Condition	Class
P ≥ 0.5	1 (Positive)
P < 0.5	0 (Negative)

☞ Default cut-off = **0.5**

2. Why Cut-Off is Important
Logistic regression gives **probability**, not class directly
☞ We must convert:
Probability → Final class

Example:

Probability	Class (cut-off = 0.5)
0.8	1
0.6	1
0.4	0
0.2	0

3. Problem with Fixed Cut-Off (0.5)
Using 0.5 is **not always optimal**

Issues:

Problem	Explanation
Imbalanced data	More of one class
Different error costs	FP ≠ FN importance
Real-world risk	Some mistakes are costly

4. What is Optimal Cut-Off
Definition: Optimal cut-off is the **best threshold value that gives maximum model performance**

☞ It balances:

- Sensitivity
- Specificity

5. Effect of Changing Cut-Off

Cut-Off Value	Effect
Low (e.g., 0.3)	More positives → ↑ Sensitivity

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High (e.g., 0.7)	More negatives → ↑ Specificity

Insight:

- Lower cut-off → fewer FN, more FP
- Higher cut-off → fewer FP, more FN

6. Example (Clear Understanding)

Predicted probabilities:

Value	Probability
A	0.8
B	0.6
C	0.4
D	0.2

Case 1: Cut-Off = 0.5

Value	Class
A	1
B	1
C	0
D	0

Case 2: Cut-Off = 0.3

Value	Class
A	1
B	1
C	1
D	0

☞ More positives detected

7. How to Choose Optimal Cut-Off

Method 1: Based on Business Need

Scenario	Choice
Disease detection	Low cut-off (avoid FN)
Spam filtering	High cut-off (avoid FP)

Method 2: ROC Curve (Important Concept)

- Plot of Sensitivity vs (1 − Specificity)
- Best cut-off = point near top-left

Method 3: Youden's Index

Formula:

J = Sensitivity + Specificity − 1

☞ Choose cut-off that maximizes J

8. Trade-Off Concept

Metric	Focus
Sensitivity	detect positives
Specificity	avoid false positives

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☞ Cannot maximize both simultaneously

9. Real-Life Example

Loan Approval

Cut-Off	Effect
Low (0.3)	More approvals (risk ↑)
High (0.7)	Fewer approvals (safe but strict)

10. Summary Table

Concept	Meaning
Cut-off probability	threshold for classification
Default value	0.5
Optimal cut-off	best threshold
Low cut-off	high sensitivity
High cut-off	high specificity

11. Key Exam Points

- Logistic regression outputs probability
- Cut-off converts probability → class
- Default = 0.5
- Optimal cut-off depends on problem
- Trade-off between sensitivity & specificity
- ROC curve used to select cut-off

12. Concept Flow (Very Important)

Model gives Probability
↓
Choose Cut-Off Value
↓
Apply Threshold Rule
↓
Convert to Class (0 or 1)
↓
Evaluate Performance
↓
Adjust Cut-Off if needed

✔ **Final Understanding**

☞ Logistic regression is not complete without choosing cut-off

☞ Correct cut-off leads to:

- better decisions
- fewer errors
- improved model performance

Variable Selection in Logistic Regression

This topic is part of Unit-6 (Logistic Regression) in your syllabus

It focuses on **selecting the most important input variables (features)** for building an effective model.

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1. What is Variable Selection

Definition: Variable selection is the process of **choosing the most relevant independent variables (X)** that significantly affect the output (Y).

Goal:

- Improve model accuracy
- Remove unnecessary variables
- Simplify the model

2. Why Variable Selection is Needed

Problem	Explanation
Too many variables	Model becomes complex
Irrelevant features	Reduce accuracy
Multicollinearity	Variables highly correlated
Overfitting	Model memorizes data

👉 Solution: Select only important variables

3. Types of Variables

Type	Meaning
Relevant variables	Important for prediction
Irrelevant variables	No impact
Redundant variables	Duplicate information

4. Methods of Variable Selection

4.1 Forward Selection

Idea: Start with no variables → add one by one

Steps:

1. Start with empty model
2. Add variable with highest significance
3. Check improvement
4. Repeat until no improvement

Advantage:

- Simple
- Useful when few variables needed

4.2 Backward Elimination

Idea:

Start with all variables → remove one by one

Steps:

1. Start with full model
2. Remove least significant variable
3. Refit model
4. Repeat until all variables are significant

Advantage:

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- More accurate
- Considers all variables initially

4.3 Stepwise Selection
Idea: Combination of forward + backward

Steps:
1. Add variable (like forward)
2. Remove unnecessary variables (like backward)
3. Repeat

Advantage:
- Balanced approach
- More flexible

5. Criteria for Selection
Variables are selected based on:

5.1 p-value (Most Important)

Value	Decision
p < 0.05	Keep variable ✔
p > 0.05	Remove variable ✕

5.2 Likelihood Ratio Test
- Compare two models
- Check if new variable improves model

5.3 AIC (Akaike Information Criterion)

Value	Meaning
Low AIC	Better model

5.4 BIC (Bayesian Information Criterion)
- Similar to AIC
- Penalizes complex models more

6. Multicollinearity Handling
Problem: Variables are highly correlated

Effect:
- Unstable coefficients
- Poor interpretation

Solution:

Method	Meaning
Remove variable	eliminate redundancy
VIF check	detect multicollinearity

7. Example (Simple Understanding)
Problem: Predict loan approval
Variables:

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- Income
- Age
- Credit Score
- Number of loans

After selection:
Keep:
- Income ✔
- Credit Score ✔

Remove:
- Age ✕
- Number of loans ✕

👉 Only important variables remain

8. Benefits of Variable Selection

Benefit	Explanation
Simpler model	easy to understand
Faster computation	less data
Better accuracy	removes noise
Better interpretation	clear relationships

9. Risks of Poor Selection

Issue	Result
Too many variables	overfitting
Too few variables	underfitting
wrong variables	incorrect predictions

10. Summary Table

Method	Approach
Forward	add variables
Backward	remove variables
Stepwise	add + remove

11. Key Exam Points
- Variable selection improves model performance
- Forward, backward, stepwise methods
- p-value is most common criteria
- AIC/BIC used for model comparison
- Avoid multicollinearity

12. Concept Flow (Very Important)
Start with Data
↓
Select Variables
↓
Build Model
↓
Evaluate Significance

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↓
Add/Remove Variables
↓
Final Optimized Model

✔ **Final Understanding**

☞ Logistic regression works best when:

- only relevant variables are used
- unnecessary variables are removed

☞ Variable selection ensures:

- accuracy
- efficiency
- interpretability

Application of Logistic Regression in Credit Rating

This topic is part of Unit-6 (Logistic Regression) in your syllabus
It explains how logistic regression is used to **decide whether a loan applicant is creditworthy or risky**.

1. What is Credit Rating

Definition: Credit rating is the process of **evaluating the ability of a customer to repay a loan**.

Output:

Class	Meaning
1	Approved (Good Credit)
0	Rejected (Bad Credit)

☞ This is a **binary classification problem**

2. Why Logistic Regression is Used

Reason	Explanation
Binary output	approve / reject
Probability output	risk estimation
Simple and interpretable	easy for banks
Fast computation	real-time decisions

3. Input Variables (Features)

Banks use multiple factors:

Variable	Meaning
Income	earning capacity
Credit score	past repayment behavior
Age	financial stability
Loan amount	risk level
Employment status	job stability
Existing loans	current liabilities

4. Model Structure

Step 1: Linear Combination

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$$z = b_0 + b_1X_1 + b_2X_2 + \cdots + b_nX_n$$

Step 2: Apply Sigmoid Function

$$P(\text{Approval}) = \frac{1}{1 + e^{-z}}$$

Step 3: Decision Rule

Probability	Decision
≥ Cut-off (e.g., 0.5)	Approve
< Cut-off	Reject

5. Example (Clear Understanding)

Suppose:

- Income = High
- Credit Score = Good

Model gives: P = 0.85

☞ Decision: 0.85 ≥ 0.5 → Approve ✔

Another case: P = 0.3

☞ Decision: 0.3 < 0.5 → Reject ✕

6. Role of Probability

Probability Range	Interpretation
0.8 – 1.0	very low risk
0.5 – 0.8	moderate risk
0.0 – 0.5	high risk

☞ Helps bank assess **risk level**

7. Benefits of Logistic Regression in Credit Rating

Benefit	Explanation
Risk prediction	estimate default probability
Objective decisions	data-driven
Transparency	easy to explain
Speed	fast approval process
Automation	reduces manual work

8. Handling Errors

Important in banking:

Error Type	Meaning	Impact
False Positive	approve risky customer	financial loss
False Negative	reject good customer	lost opportunity

☞ Banks adjust cut-off to balance risk

9. Role of Cut-Off Probability

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Cut-Off	Effect
Low (0.3)	more approvals (higher risk)
High (0.7)	fewer approvals (safer)

☞ Banks usually prefer **higher cut-off** for safety

10. Model Evaluation

Banks use:

- Confusion matrix
- Sensitivity & Specificity
- ROC curve

☞ To ensure reliable predictions

11. Real-World Workflow

Customer applies for loan
↓
Collect data (income, credit score, etc.)
↓
Apply logistic regression model
↓
Calculate probability of approval
↓
Compare with cut-off
↓
Approve / Reject decision

12. Summary Table

Concept	Meaning
Problem type	binary classification
Output	approve / reject
Model output	probability
Decision	based on cut-off
Goal	minimize risk

13. Key Exam Points

- Logistic regression predicts **probability of default/approval**
- Used in banking and finance
- Helps in **risk management**
- Cut-off selection is critical
- Errors have financial impact

✔ **Final Understanding**

☞ Logistic regression helps banks answer:

- Will customer repay loan?
- How risky is the customer?

☞ It converts raw customer data into:

- probability
- decision
- business action

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Gain Chart and Lift Chart

This topic is part of Unit-6 (Logistic Regression) in your syllabus
It is used to **evaluate and compare classification models**, especially in business applications like marketing and credit scoring.

1. Why Gain & Lift Charts are Needed

Basic metrics (accuracy, sensitivity) are not enough when:

- Data is imbalanced
- Business wants **targeted decisions**
- Need to rank customers based on probability

☞ Gain & Lift charts evaluate **ranking quality of model**

2. Gain Chart

Definition: Gain chart shows **how much better the model performs compared to random selection**

Idea

- Sort data by predicted probability (high → low)
- Divide into groups (deciles: top 10%, 20%, ...)
- Measure how many actual positives captured

Interpretation

Percentage of Data	Gain
Top 10%	captures most positives
Top 50%	captures majority
100%	captures all

☞ Good model:

- Steep curve
- High gain in early %

Example

If:

- Top 20% customers contain 60% defaulters

☞ Model is very effective

3. Lift Chart

Definition: Lift chart measures **how much better the model is compared to random guessing**

Formula:

Lift = (Model Performance) / (Random Performance)

Interpretation

Lift Value	Meaning
= 1	No improvement
> 1	Good model
< 1	Poor model

Example

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If Lift = 3 at 20% data:
👉 Model is **3 times better than random**

4. Gain vs Lift

Feature	Gain Chart	Lift Chart
Focus	cumulative performance	relative improvement
Output	% positives captured	improvement factor
Use	targeting strategy	model comparison

5. Real-Life Use Case
Marketing Campaign

- Select top 20% customers
- Gain chart → shows % responders
- Lift chart → shows improvement

👉 Helps maximize profit

6. Key Insight

- Gain → "How much we captured"
- Lift → "How much better we are"

7. Summary Table

Concept	Meaning
Gain chart	cumulative success
Lift chart	improvement over random
Good model	high gain, high lift
Use	ranking & targeting

8. Concept Flow
Model predicts probabilities
↓
Sort data (high → low)
↓
Divide into groups
↓
Calculate gain & lift
↓
Evaluate model

✔ **Final Understanding**
👉 These charts are **business-focused evaluation tools**
👉 Used in:

- credit scoring
- marketing
- fraud detection

🔥 **Other Important Charts Used in Data Science**

1. ROC Curve (Very Important)

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Definition: ROC (Receiver Operating Characteristic) curve shows:
Sensitivity vs (1 − Specificity)

Interpretation

Curve	Meaning
Closer to top-left	good model
Diagonal line	random model

AUC (Area Under Curve)

Value	Meaning
1	perfect model
0.5	random

2. Precision-Recall Curve
Definition: Shows relationship between:

- Precision
- Recall (Sensitivity)

Use

- Best for **imbalanced datasets**

3. Confusion Matrix (Basic Chart)
Already covered:

- TP, TN, FP, FN
- Base for all metrics

4. Histogram
Purpose: Shows distribution of data
Use

- understand spread
- detect skewness

5. Box Plot
Purpose : Shows:

- median
- quartiles
- outliers

6. Scatter Plot
Purpose: Shows relationship between variables

7. Line Chart
Purpose: Trend over time

8. Bar Chart
Purpose: Compare categories

9. Heatmap
Purpose: Shows correlation between variables

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10. Decision Boundary Plot
Purpose: Visualize classification regions

📊 **Final Summary Table**

Chart	Purpose
Gain Chart	cumulative performance
Lift Chart	improvement vs random
ROC Curve	classification performance
Precision-Recall	imbalanced data
Confusion Matrix	error analysis
Histogram	distribution
Box Plot	outliers
Scatter Plot	relationships
Heatmap	correlation

✔ **Final Understanding**
👉 In Logistic Regression:

- Confusion Matrix → basic evaluation
- Sensitivity/Specificity → detailed metrics
- ROC → threshold performance
- Gain/Lift → business decision quality

👉 Together, they give **complete model evaluation**

Logistic Regression Model Evaluation

This topic summarizes how to **evaluate the performance of a logistic regression model** using standard metrics.

1. What is Model Evaluation
Definition: Model evaluation measures **how well the model predicts correct classes**.
👉 It answers:

- Is the model accurate?
- How many errors does it make?
- What type of errors?

2. Confusion Matrix (Base of All Metrics)

Actual \ Predicted	1 (Positive)	0 (Negative)
1 (Positive)	TP	FN
0 (Negative)	FP	TN

Terms:

Term	Meaning
TP	Correct positive
TN	Correct negative
FP	Wrong positive
FN	Wrong negative

3. Accuracy
Definition: Accuracy measures **overall correctness of the model**

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Formula:
Accuracy = (TP + TN) / (TP + TN + FP + FN)

Interpretation:

Value	Meaning
High accuracy	Good model
Low accuracy	Poor model

Limitation:

- Not reliable for **imbalanced data**

4. Precision
Definition : Precision measures **how many predicted positives are actually correct**

Formula: Precision = TP / (TP + FP)

Interpretation:

Value	Meaning
High precision	Few false positives
Low precision	Many false positives

Example:
Spam detection
☞ High precision = fewer normal emails marked as spam

5. Recall (Sensitivity)
Definition: Recall measures **how many actual positives are correctly identified**

Formula: Recall = TP / (TP + FN)

Interpretation:

Value	Meaning
High recall	Few missed cases
Low recall	Many missed cases

Example:
Disease detection
☞ High recall = fewer patients missed

6. F1-Score
Definition: F1-score is the **harmonic mean of precision and recall**

Formula: F1 = 2 × (Precision × Recall) / (Precision + Recall)

Why Needed:
Balances:

- Precision
- Recall

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Interpretation:

Value	Meaning
High F1	balanced performance
Low F1	poor balance

7. Example (Complete Calculation)

Suppose:

Value	Count
TP	50
TN	40
FP	10
FN	20

Step 1: Accuracy

Accuracy = (50 + 40) / 120 = 0.75

Step 2: Precision

Precision = 50 / (50 + 10) = 0.83

Step 3: Recall

Recall = 50 / (50 + 20) = 0.71

Step 4: F1-score

F1 ≈ 0.77

8. Key Differences

Metric	Focus	Best Use
Accuracy	overall	balanced data
Precision	FP reduction	spam filtering
Recall	FN reduction	medical diagnosis
F1-score	balance	imbalanced data

9. When to Use What

Situation	Metric
Equal class distribution	Accuracy
Avoid false positives	Precision
Avoid false negatives	Recall
Need balance	F1-score

10. Summary Table

Metric	Formula
Accuracy	(TP + TN) / Total
Precision	TP / (TP + FP)
Recall	TP / (TP + FN)
F1-score	2PR / (P + R)

11. Concept Flow (Very Important)

Model Prediction

↓

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Build Confusion Matrix
↓
Calculate Metrics
↓
Compare Results
↓
Evaluate Model

✔ **Final Understanding**

👉 Confusion matrix is the base

👉 Metrics give deeper insight

👉 Good model depends on problem:

- Not just accuracy
- Balance between errors

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